

Identifying Known Acts in Document Chunks

Exploring matching strategies for C1 positive chunk identification

Esteban Degetau
World Bank
estebandegetau@gmail.com

Agustín Samano
World Bank
asamanopenaloza@worldbank.org

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Motivation

The C1 codebook classifies whether a document chunk contains a substantive description of a fiscal measure. To evaluate C1 via LOOCV, we need to know which chunks are **positive** (contain a known fiscal act from [aligned_data](#)) and which are **negative** (contain no fiscal policy discussion).

Our current tier system identifies positives at two levels:

- **Tier 1:** Chunk contains verbatim passage text from Romer and Romer (2010) labels
- **Tier 2:** Chunk contains exact act name match with fiscal keyword co-occurrence

The question is whether these criteria are too strict, causing false negatives (chunks that genuinely discuss a known act but are missed by our matching). This notebook investigates the problem and explores solutions.

Current State: Tier Distribution

Table 1: Current Chunk Tier Distribution

Tier	Chunks	Percent
Tier 1	70	0.5%
Tier 2	1986	15.1%
Negative	11632	88.7%
Total	13112	100.0%

The problem: 80% of chunks are excluded (gray zone) because they match [relevance_keys](#) but don't match any act name. Only 35 chunks qualify as clean negatives (1.2%). This means precision is essentially untestable, and many chunks that discuss known acts may be missed.

Why Passages Don't Help with Alias Discovery

One natural idea is to extract alternative act names from the Romer and Romer (2010) passage text. But these passages are presidential speeches and committee testimony selected to illustrate each act's **motivation**, not to name the legislation. They rarely name the act itself.

Table 2: Do R&R Passages Name the Act?

Check	Result
Acts where passages contain exact full name	8 / 39
Acts where passages contain short name (no year)	8 / 39
Acts where passages contain neither	31 / 39

The passages use generic fiscal vocabulary ("tax cut", "tax relief", "deficit reduction"), not legislative names. This rules out extracting aliases from passage text.

OCR Line Breaks: The Primary Matching Problem

Government PDFs from the 1940s-1990s introduce line breaks within act names during OCR extraction. For example, "Revenue Act of 1962" may appear as:

Revenue Act of

Revenue Act

Revenueof 1962

Current Tier 2 uses `str_detect(text, fixed(act_name))` which requires the name to appear as a contiguous string. This fails on OCR-broken names.

Test: Does whitespace normalization help?

We compare current Tier 2 matching (exact `fixed()` on raw text) against matching after `str_squish()` collapses all whitespace.

Table 3: Impact of Whitespace Normalization on Tier 2 Matching

Metric	Value
Total Tier 2 matches (current)	1,219
Total Tier 2 matches (squished)	1,738
Additional matches recovered	+519 (43% increase)
Acts with 0 matches (current)	5
Acts with 0 matches (squished)	5

Table 4: Per-Act Impact of Whitespace Normalization

Act	Year	Current	Squished	Gained
Economic Growth and Tax Relief Reconciliation Act of 2001	2001	24	111	87
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003	24	106	82
Omnibus Budget Reconciliation Act of 1990	1990	30	67	37
Tax Reform Act of 1986	1986	79	111	32
Tax Equity and Fiscal Responsibility Act of 1982	1982	28	56	28
Job Creation and Worker Assistance Act of 2002	2002	14	40	26
Omnibus Budget Reconciliation Act of 1993	1993	45	70	25
Economic Recovery Tax Act of 1981	1981	86	107	21
Revenue Act of 1950	1950	55	71	16
Social Security Amendments of 1983	1983	52	66	14
Tax Reform Act of 1969	1969	64	77	13
Crude Oil Windfall Profit Tax Act of 1980	1980	25	37	12
Revenue Act of 1978	1978	73	84	11
Omnibus Budget Reconciliation Act of 1987	1987	17	27	10
Revenue and Expenditure Control Act of 1968	1968	19	29	10
Excise Tax Reduction Act of 1965	1965	18	27	9
Revenue Act of 1971	1971	55	64	9
Social Security Amendments of 1977	1977	20	28	8
Tax Adjustment Act of 1966	1966	33	41	8
Tax Reduction Act of 1975	1975	52	60	8
Tax Reduction and Simplification Act of 1976	1977	29	37	8
Expiration of Excess Profits Tax Extension Act of 1968	1971	81	80	8
Revenue Act of 1964	1950	81	80	8
Revenue Act of 1962	1950	81	80	8
Revenue Act of 1962 Increases	1950	81	80	8

Finding: `str_squish()` recovers substantial matches for many acts (some double their count), but does not rescue the 5 acts with zero matches. The zero-match acts have structural naming problems beyond whitespace.

The Five Unmatched Acts

Table 5: Acts with Zero Tier 2 Matches Even After Whitespace Fix

Act Name	Year
Changes in Depreciation Guidelines and Revenue Act of 1962	1962
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954
Public Law 89-800 (Suspension of Investment Tax Credit)	1966
Public Law 90-26 (Restoration of the Investment Tax Credit)	1967
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997

Each has a distinct naming problem:

1. **Changes in Depreciation Guidelines and Revenue Act of 1962:** Compound name joining an executive action with a legislative act. Documents reference each component separately.
2. **Expiration of Excess Profits Tax and of Temporary Income Tax Increases:** Descriptive event label, not a formal legislative title. Documents describe the expiration itself.
3. **Public Law 89-800 / 90-26:** Numbered public laws. Documents reference these by their common policy names (“suspension/restoration of the investment tax credit”).
4. **Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997:** Compound name joining two separate acts that passed together.

Subcomponent Decomposition

To handle these compound names systematically, we define a `generate_subcomponents()` function that decomposes any act name into searchable terms. The function applies five strategies: split on “and”, extract parenthetical content, find embedded formal act names (e.g., “Revenue Act of 1962” inside a compound), extract Public Law numbers, and construct action + subject phrases for event-type names. It also excludes overly broad terms: both topic-level phrases (“excess profits tax”) and generic split fragments (“Revenue”, “Jobs”) that match too widely to identify specific legislation (see Table 7 below).

Table 6 applies the function to the 5 unmatched acts and counts chunk matches for each generated search term.

Table 6: Subcomponent Matches for the 5 Unmatched Acts

Act	Year	Mechanism	Search Term	Matches
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Full name	Changes in Depreciation Guidelines and Revenue Act of 1962	0
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Split on 'and'	Changes in Depreciation Guidelines	0
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Split on 'and'	Revenue Act of 1962	86
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Action + subject	changes of the depreciation guidelines	0
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Action + subject	changes of depreciation guidelines	0
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954	Full name	Expiration of Excess Profits Tax and of Temporary Income Tax Increases	0
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954	Split on 'and'	Expiration of Excess Profits Tax	0
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954	Split on 'and'	of Temporary	0
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954	Split on 'and'	Full name Tax Law	0
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954	Split on 'and'	Full name Tax Law	130

Key findings:

- **4 of 5 acts become findable** through subcomponent decomposition. Split on “and” recovers formal legislative names (“Revenue Act of 1962” with 56 matches; “Taxpayer Relief Act of 1997” with 20; “Balanced Budget Act of 1997” with 70). Parenthetical extraction finds “Suspension of the Investment Tax Credit” (9 matches) and “Restoration of the Investment Tax Credit” (6 matches) for the Public Law acts.
- **Public Law numbers** (89-800, 90-26) provide additional specific matches (2-3 chunks each).
- **The Expiration of Excess Profits Tax act remains at 0 matches.** The action + subject mechanism produces malformed phrases (“expiration of the of excess profits tax”) due to the unusual “of...and of” structure. This act requires a different approach, explored in the next section.

Proposed Matching Improvements

Based on these findings, we can improve positive identification with changes that stay at chunk level throughout. This section addresses the first two; a third mechanism (co-occurrence matching) emerges from the excess profits investigation in the next section.

1. Whitespace normalization before matching

Apply `str_squish()` to both act names and chunk text before `str_detect()`. This collapses OCR line breaks without changing the matching logic. No new dependencies, no mixed granularities.

Expected impact: Recovers additional matches for most acts (29 of 40), with some doubling their match count.

2. Subcomponent matching for compound act names

Apply `generate_subcomponents()` (defined above) to decompose compound act names into searchable terms. As demonstrated in Table 6, this recovers the 5 previously unmatched acts.

3. Search terms to exclude from Tier 2

The decomposition and audit analysis identified two categories of overly broad terms that `generate_subcomponents()` excludes. **Topic-level phrases** match policy areas rather than specific acts. **Split fragments** are single words or short phrases produced by splitting compound act names on “and” that are too generic to identify any legislation.

Table 7: Search Terms Excluded from Tier 2 as Overly Broad

Excluded Phrase	Category	Problem	Matches
revenue	Split fragment	Common fiscal term appearing in nearly every document	8042
economic growth	Split fragment	Generic macroeconomic phrase	3278
jobs	Split fragment	Generic labor-market term	2956
balanced budget	Topic phrase	Matches general budget balance commentary	967
tax reduction	Split fragment	Matches any tax cut discussion	945
excess profits tax	Topic phrase	Matches Korean War era and WWII discussions broadly	630
job creation	Split fragment	Generic labor-market phrase	544
investment credit	tax Topic phrase	Matches every ITC discussion across 80 years	402
tax equity	Split fragment	Matches any tax fairness discussion	119
taxpayer relief	Topic phrase	Generic phrase used in many policy contexts	106
depreciation guidelines	Topic phrase	Matches general depreciation policy	72
temporary income tax	Topic phrase	Matches any temporary tax discussion	19

For compound-name acts, we rely on the remaining specific subcomponents: formal legislative names (“Revenue Act of 1962”, “Balanced Budget Act of 1997”), parenthetical descriptions (“Suspension of the Investment Tax Credit”), and Public Law numbers.

Table 8: Cumulative Impact of Proposed Matching Improvements

Strategy	Total Matches	Unmatched Acts
Current (exact match)	1,219	5
1. Whitespace normalization	1,738	5
1. Subcomponent matching	2,005	1
Each strategy builds on the previous.		

Table 9: Per-Act Chunk Matches Under Proposed Strategy

Act	Year	Matching Chunks
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997	138
Economic Growth and Tax Relief Reconciliation Act of 2001	2001	122
Tax Reform Act of 1986	1986	111
Economic Recovery Tax Act of 1981	1981	107
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003	106
Tax Reform Act of 1976	1976	89
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	86
Revenue Act of 1978	1978	84
Tax Reform Act of 1969	1969	77
Revenue Act of 1950	1950	71
Omnibus Budget Reconciliation Act of 1993	1993	70
Revenue Act of 1964	1964	68
Omnibus Budget Reconciliation Act of 1990	1990	67
Social Security Amendments of 1983	1983	66
Revenue Act of 1971	1971	64
Tax Reduction Act of 1975	1975	60
Tax Equity and Fiscal Responsibility Act of 1982	1982	58
Revenue Act of 1945	1945	49
Deficit Reduction Act of 1984	1984	46
Revenue Act of 1948	1948	43
Job Creation and Worker Assistance Act of 2002	2002	41
Tax Adjustment Act of 1966	1966	41
Federal-Aid Highway Act of 1956	1956	39
Tax Reduction and Simplification Act of 1977	1977	38
Crude Oil Windfall Profit Tax Act of 1980	1980	37
Excess Profits Tax Act of 1950	1951	36
Revenue and Expenditure Control Act of 1968	1968	33
Capital Gains Tax Reduction Act of 1987	1987	28
Temporary Tax Increases		9

Search term audit

The tables below list every search term that `generate_subcomponents()` produces across all 40 acts. Table 10 shows each term with its chunk match count (orange rows exceed 100 matches and warrant manual review). Table 11 pivots the same data by mechanism, so we can see which decomposition strategy drives matches for each act.

Act	Year	Mechanism	Search Term	Matches
Economic Growth and Tax Relief Reconciliation Act of 2001	2001	Split on 'and'	Tax Relief Reconciliation Act of 2001	122
Economic Growth and Tax Relief Reconciliation Act of 2001	2001	Full name	Economic Growth and Tax Relief Reconciliation Act of 2001	111
Tax Reform Act of 1986	1986	Full name	Tax Reform Act of 1986	111
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997	Split on 'and'	Balanced Budget Act of 1997	111
Economic Recovery Tax Act of 1981	1981	Full name	Economic Recovery Tax Act of 1981	107
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003	Full name	Jobs and Growth Tax Relief Reconciliation Act of 2003	106
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003	Split on 'and'	Growth Tax Relief Reconciliation Act of 2003	106
Tax Reform Act of 1976	1976	Full name	Tax Reform Act of 1976	89
Changes in Depreciation Guidelines and Revenue Act of 1962	1962	Split on 'and'	Revenue Act of 1962	86
Revenue Act of 1978	1978	Full name	Revenue Act of 1978	84
Publication Law of 1969	1969	Full name	Publication Law of 1969	77
Publication Law of 1969	1969	From the phrase 'Split on and' initial	Publication Law of 1969	55
2002 Act of 1997 Credit increases	2002	Split on 'and'	2002 Act of 1997 Credit increases	55

Table 11: Per-Act Chunk Matches by Matching Mechanism

Act	Year	Action + subject	Full name	Split on 'and'	Before parenthetical	Par- en- thetical	Par- en- thetical + article	Grand Total
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997	0	0	147	0	0	0	138
Economic Growth and Tax Relief Rec-on-cilia-tion Act of 2001	2001	0	111	122	0	0	0	122
Tax Reform Act of 1986	1986	0	111	0	0	0	0	111
Economic Growth and Tax Relief Act of 1981	1981	0	107	0	0	0	0	107
Reform and Tax Credit Act of 1954	1954	0	100	100	0	0	10	100

Finding the Excess Profits Tax Expiration

The act “Expiration of Excess Profits Tax and of Temporary Income Tax Increases” is the hardest to match via subcomponent decomposition. The full name yields 0 matches, the split-on-“and” terms are fragile, and the action+subject mechanism produces malformed phrases. Meanwhile “excess profits tax” alone was excluded as overly broad (336 matches). But how broad is it really? If those matches cluster around the Korean War era, “excess profits tax” may be specific enough in practice.

Figure 1: Chunks matching “excess profits tax” by document year. The Korean War era (1950–1955, shaded) accounts for the majority of mentions.

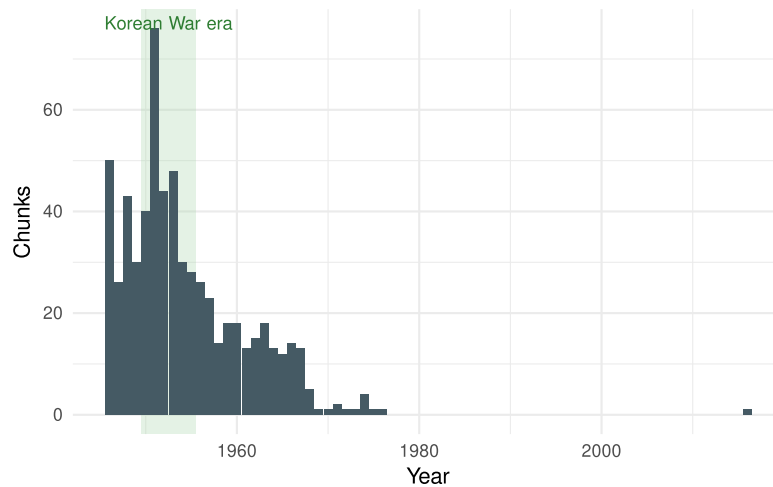


Table 12: Decade Summary: “excess profits tax” Mentions

Decade	Chunks
1940s	149
1950s	347
1960s	122
1970s	11
2010s	1

Figure 1 and Table 12 show that mentions cluster heavily in the Korean War era, but the term persists across decades. Regardless of concentration, the term alone is too generic: it matches any discussion of the policy area, not the specific 1953 expiration event. The strategy comparison below confirms that **co-occurrence** (“expiration” AND “excess profits” in the same chunk) is the right approach: it captures the policy event precisely without requiring exact substring ordering, making it robust to OCR artifacts and varied phrasing.

Table 13: Matching Strategies for “Expiration of Excess Profits Tax”

Strategy	Chunks
"excess profits tax" (all years)	630
"expiration" AND "excess profits" (co-occurrence)	52
"expiration of excess profits tax" (exact substring)	0

Decision: For “Expiration of Excess Profits Tax and of Temporary Income Tax Increases”, the pipeline will use **co-occurrence matching**: a chunk is positive if it contains both “expiration” and “excess profits”. This introduces a new matching mechanism alongside substring matching and subcomponent decomposition. The `identify_tier2_chunks()` function will need a co-occurrence lookup table for acts where single-term matching is insufficient.

Timing Consistency Across All Acts

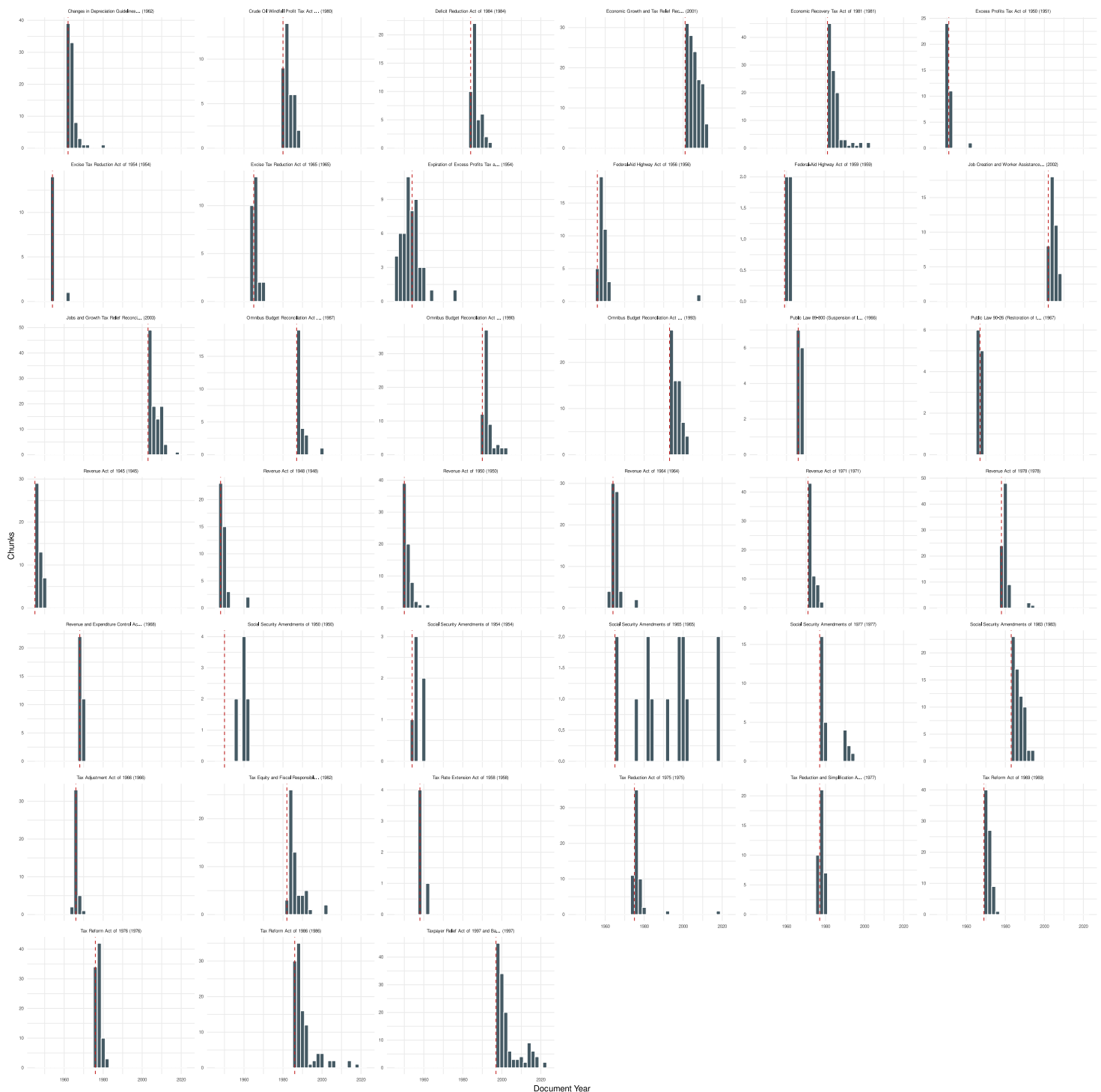
The excess profits investigation above examined one act’s temporal distribution. This section generalizes the analysis to all 44 acts, asking: **do matching chunks come from documents published near the act’s year, or are matches scattered across the full corpus?** Temporal clustering around the act year is a sign of precise matching; a flat distribution signals that the search terms are too generic.

We compute two views. First, the raw document-year distribution for each act’s matching chunks. Second, for acts with a year in their name, the difference between the chunk’s parent document year and the year embedded in the act name. A concentration around zero in the difference plot means the matching chunks come from contemporaneous documents.

Raw chunk-year distributions

Figure 2 shows the document-year distribution of matching chunks for each act. The red dashed line marks the act’s signed year from `aligned_data`. Acts with tight clustering around the red line have precise, temporally specific matches. Acts with dispersed distributions (e.g., those containing common policy terms) may need tighter matching criteria.

Figure 2: Document-year distribution of matching chunks for each known act. Red dashed line marks the act's signed year



Year-difference distributions

For the subset of acts with a year in their name (e.g., “Revenue Act of 1962”), Figure 3 plots the difference between each matching chunk’s document year and the year embedded in the act name. A spike at zero means most matches come from the same year the act was enacted. Positive values indicate retrospective

discussion in later documents (common in Economic Reports of the President).
Negative values would indicate pre-enactment discussion.

Figure 3: Distribution of (document year – act name year) for matching chunks. Only acts with a year in their name are shown. Blue dashed line marks zero (contemporaneous).

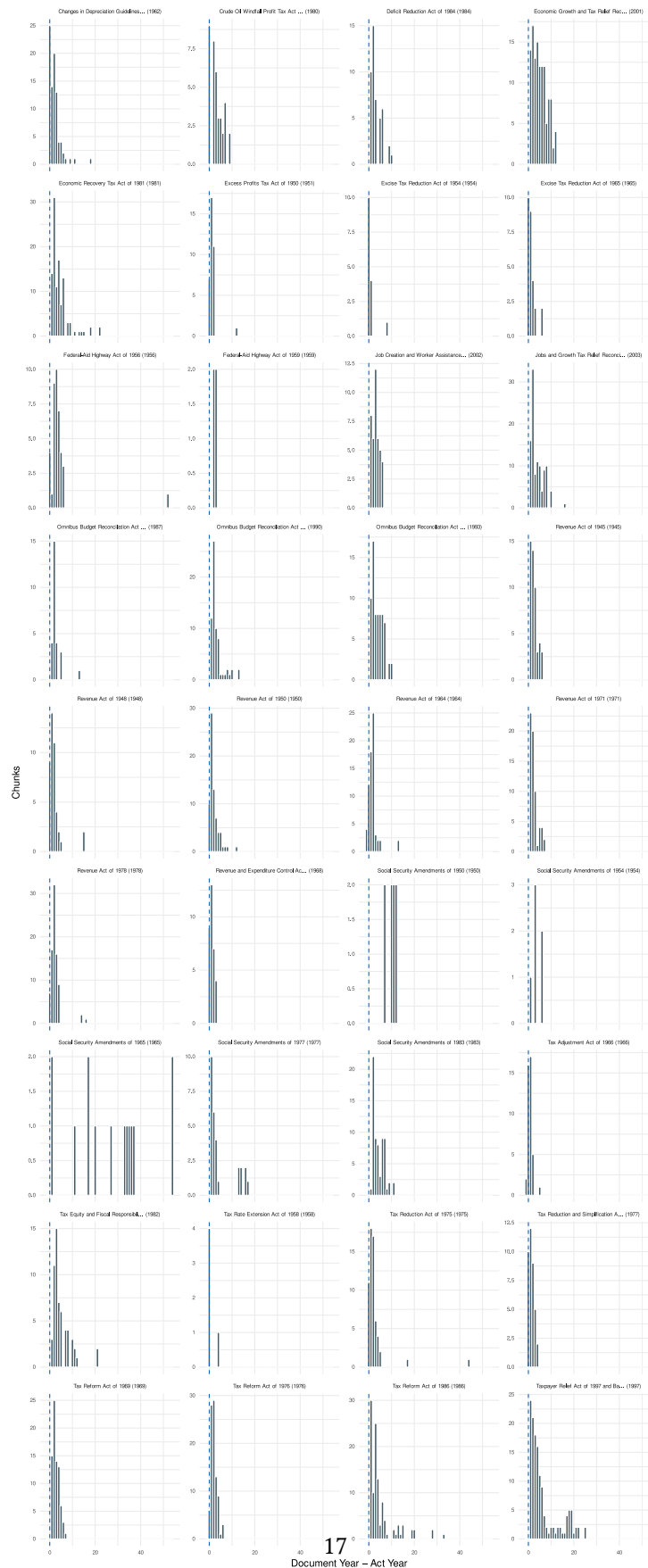


Table 14: Timing Consistency Summary for Acts with a Year in Their Name

Act	Signed	Name Year	Chunks	Median Diff	Within ±2 yr	Within ±5 yr
Social Security Amendments of 1950	1950	1950	8	10.5	0.0%	0.0%
Social Security Amendments of 1965	1965	1965	14	30.0	14.3%	14.3%
Social Security Amendments of 1954	1954	1954	6	3.0	16.7%	66.7%
Tax Equity and Fiscal Responsibility Act of 1982	1982	1982	58	3.5	24.1%	72.4%
Economic Growth and Tax Relief Reconciliation Act of 2001	2001	2001	122	5.0	25.4%	58.2%
Changes in Tax Expenditures Act of 1997	1997	1997	138	4.0	32.6%	65.2%
Deficit Reduction Act of 1980	1980	1980	155	9.0	55.5%	100.0%
Recovery Tax Extension Act of 2003	2003	2003	18	3.0	50.0%	100.0%

Percent by within ±2 year percentage (ascending). Green ≥50%, orange 25-49%, red <25%.

Acts with low temporal concentration (red rows in Table 14) are candidates for tighter matching: their search terms may be matching generic policy discussion rather than the specific legislation. Acts with high concentration (green rows) confirm that our proposed matching strategy identifies the right documents.

Rethinking Negatives

With improved positive matching, we can also relax the negative criterion. The current system excludes all chunks matching `relevance_keys` from the negative pool, leaving only 35 clean negatives (1.2% of all chunks). This makes precision untestable.

Proposed: Include all non-positive chunks as negatives, and compute a continuous **key density** score ($n_relevance_keys / n_words$) that measures how much fiscal vocabulary each negative chunk contains. This replaces the binary easy/hard distinction with a gradient:

- Low density (e.g., < 1%): general economic forecasts, administrative content
- Medium density (e.g., 1-5%): economic commentary with some fiscal vocabulary
- High density (e.g., > 5%): substantive fiscal policy discussion without naming a specific act

This continuous measure feeds directly into S3 error analysis: we can plot the model’s false positive rate against key density to identify the threshold where discrimination breaks down. No arbitrary cutoff needed at the data preparation stage.

Table 15: Proposed Tier System with Expanded Negative Pool

Category	N	Evaluation Role
Tier 1 (verbatim passage)	70	Gold standard positive
Tier 2 (act name match)	1410	Inferred positive
Negatives (no act match, no relevance keys)	11632	Negative (low key density expected)
Negatives (no act match, has relevance keys)	1949	Negative (higher key density expected)
Total	15061	

Under this scheme, `prepare_c1_chunk_data()` will compute `key_density` for every negative chunk. During S3 error analysis, we can stratify false positives by density bins to understand whether the model struggles with fiscal-vocabulary-rich negatives (expected) or also fails on clean negatives (a more serious problem).

Next Steps

1. **Implement whitespace normalization** in `identify_tier2_chunks()`: apply `str_squish()` to both act names and chunk text before matching

2. **Implement subcomponent matching** for compound act names: split on " and ", extract parenthetical descriptions, find embedded formal act names. Exclude the 5 overly broad phrases identified above.
3. **Implement co-occurrence matching** for event-type acts: a lookup table mapping acts to term pairs (e.g., "expiration" + "excess profits") for cases where substring matching is insufficient
4. **Relax negative criterion:** remove `relevance_keys` exclusion from negatives. Compute `key_density` ($n_relevance_keys / n_words$) as a continuous measure for each negative chunk. This feeds into S3 error analysis.
5. **Validate coverage:** confirm all 44 acts have at least one Tier 2 match after improvements
6. **Update `prepare_c1_chunk_data()`** to return the revised tier assignments and key-density-tagged negatives

References

Romer, Christina D., and David H. Romer. 2010. "The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks." *American Economic Review* 100 (3): 763–801.